Madan Bhandari Memorial College

Lab No. 1

Write a C program to find the growth in national consumption for five years using Distributed Lag Model given below:

$I = 2 + 0.1\ Y\text{-}1$

$Y = 45.45 + 2.27\ (I + G)$

$T = 0.2\ Y$

$C = 20 + 0.7\ (Y - T)$

Assume the initial value of Y_{-1} is 80 and take the governmental expenditure in the 5 years to be as follows:

Year	G
1	20
2	25
3	30
4	35
5	40

Source Code:

```
#include <stdio.h>
#include <conio.h>
int main()
{
   float Y_1, Y, I, T, C[5];
   float G[5] = {20, 25, 30, 35, 40};
   int i;
   printf("Enter initial value of lagged variable Y_1:");
   scanf("%f", &Y_1);
   printf("\nThe growth in consumption is given following tables:\n");
   printf("\nYear \t \t Consumption\n");
   for (i = 0; i < 5; i++)
   {
      I = 2 + 0.1 * Y_1;
      Y = 45.45 + 2.27 * (I + G[i]);
      T = 0.2 * Y;
      C[i] = 20 + 0.7 * (Y - T);
      printf("\n %d \t \t %.2f", i + 1, C[i]);
      Y_1 = Y;
   }
   return 0;
}
```

Output:

```
C:\csit\5_fifth sem\jonash sim

Enter initial value of lagged variable Y_1:80

The growth in consumption is given following tables:

Year               Consumption

 1                 83.59
 2                 94.21
 3                 102.98
 4                 111.32
 5                 119.57
--------------------------------
Process exited after 2.354 seconds with return value 0
Press any key to continue . . .
```

Madan Bhandari Memorial College

Lab No. 2

Customers arrive in a bank according to a Poisson's process with mean inter arrival time of 10 minutes. Customers spend an average of 5 minutes on the single available counter, and leave.

Write a program in C to find:

I. Probability that a customer will not have to wait at the counter.

II. Expected number of customers in the bank.

III. Time can a customer expect to spend in the bank

Source Code:

```c
#include <stdio.h>
#include <conio.h>
int main()
{
   float l, m, e;
   float p, et;
   printf("Enter Inter arrival time of customers per hours:");
   scanf("%f", &l);
   printf("\nEnter The average time spent by cutomers per hour:");
   scanf("%f", &m);
   p = 1 - l / m;
   e = l / (m - l);
   et = 1 / (m - l);
   et = et * 60;
   printf("\nThe probability That a customer wont't have to wait at counter:%f ", p);
   printf("\n\nExpected number of customer: %f", e);
   printf("\n\nExpected time to be spent in bank: %f minutes", et);
   return 0;
}
```

Output:

```
C:\csit\5_fifth sem\jonash sim

Enter Inter arrival time of customers per hours:10

Enter The average time spent by cutomers per hour:15

The probability That a customer wont't have to wait at counter:0.333333

Expected number of customer: 2.000000

Expected time to be spent in bank: 12.000000 minutes
--------------------------------
Process exited after 20.17 seconds with return value 0
Press any key to continue . . .
```

Madan Bhandari Memorial College

Lab No. 3

At the ticket counter of football stadium, people come in queue and purchase tickets. Arrival rate of customers is 1/min. It takes at the average 20 seconds to purchase the ticket.

WAP in C to calculate total time spent by a sports fan to be seated in his seat, if it takes 1.5 minutes to reach the correct seat after purchasing the ticket. if a fan comes exactly before 2 minutes before the game starts, can sports fan expects to be seated for the kick-off?

Source Code:

```c
#include <stdio.h>

int main()
{
   float arrival_rate = 1.0;
   float service_time_sec = 20.0;
   float service_rate = 60.0 / service_time_sec;
   float travel_to_seat_min = 1.5;
   float arrival_before_kickoff = 2.0;
   float time_in_system_min = 1.0 / (service_rate - arrival_rate);
   float total_time_min = time_in_system_min + travel_to_seat_min;

   printf("Total time to reach seat: %.2f minutes\n", total_time_min);

   if (total_time_min <= arrival_before_kickoff)
   {
      printf("Result: Yes, the fan will be seated before kick-off.\n");
   }
   else
   {
      printf("Result: No, the fan will miss the start of the game.\n");
   }
   return 0;
}
```

Output:

```
C:\csit\5_fifth sem\jonash sim

Total time to reach seat: 2.00 minutes
Result: Yes, the fan will be seated before kick-off.

--------------------------------
Process exited after 0.07672 seconds with return value 0
Press any key to continue . . .
```

Lab No. 4

In a single pump service station, vehicles arrive for fuelling with an average of 5 minutes between arrivals. If an hour is taken as unit of time, cars arrive according to Poison's process with an average of λ= 12 cars/hr. Write a C program to generate Poisson distribution for x = 0,1,2, …. 15.

$$f(x) = \Pr(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-12}12^x}{x!}, \begin{cases} x = 0, 1, 2, .. \\ \lambda > 0 \end{cases}$$

Source Code:

```
#include <stdio.h>
#include <conio.h>
#include <math.h>
int main()
{
  float l, m, p[16], pr, x, a, i;
  int j;
  printf("Enter arrival rate per hours:");
  scanf("%f", &l);
  for (x = 0; x < 16; x++)
  {
    a = 1;
    for (i = 1; i <= x; i++)
    {
      a = a * i;
    }
    pr = (pow(2.71, -l) * pow(l, x)) / a;
    printf("P(X=%f) = %f\n", x, pr);
  }
}
```

Output:

```
C:\csit\5_fifth sem\jonash sim

Enter arrival rate per hours:12
P(X=0.000000) = 0.000006
P(X=1.000000) = 0.000076
P(X=2.000000) = 0.000459
P(X=3.000000) = 0.001836
P(X=4.000000) = 0.005507
P(X=5.000000) = 0.013216
P(X=6.000000) = 0.026432
P(X=7.000000) = 0.045311
P(X=8.000000) = 0.067967
P(X=9.000000) = 0.090623
P(X=10.000000) = 0.108747
P(X=11.000000) = 0.118633
P(X=12.000000) = 0.118633
P(X=13.000000) = 0.109508
P(X=14.000000) = 0.093864
P(X=15.000000) = 0.075091
```

Lab No. 5

The probabilities of weather conditions (modelled as either rainy or sunny), given the weather on the preceding day, can be represented by a transition matrix:

0.9 0.1

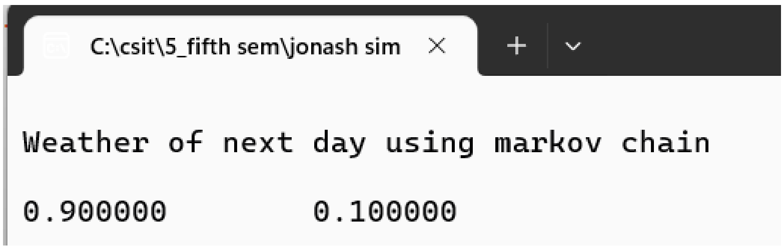
0.5 0.5

The weather on day 0 is known to be sunny. This is represented by a vector in which the "sunny" entry is 100%, and the "rainy" entry is 0%: (1 0). Write a C program to find the weather of the next day by using Markov Chain Method.

Source Code:

```c
#include <stdio.h>
#include <conio.h>
int main()
{
  float transMat[2][2] = {0.9, 0.1, 0.5, 0.5}, vect[1][2] = {1, 0}, result[1][2];
  int i, j, k;
  for (i = 0; i < 2; i++)
  {
    for (j = 0; j < 2; j++)
    {
      result[i][j] = 0;
      for (k = 0; k < 2; k++)
      {
        result[i][j] += vect[i][k] * transMat[k][j];
      }
    }
  }
  printf("\nWeather of next day using markov chain\n\n");
  for (i = 0; i < 1; i++)
  {
    for (j = 0; j < 2; j++)
    {
      printf("%f\t", result[i][j]);
    }
    printf("\n");
  }
  return 0;
}
```

Output:



MBM
Madan Bhandari Memorial College

Lab No. 6

Q.N. 1: What is random number and what are its properties. WAP in C to generate 100 random numbers using Linear Congruential Method where X0=11, m=100, a = 5 and c = 13.

Source Code:

```c
#include <stdio.h>
#include <conio.h>
#define SIZE 100
int main()
{
   int x[SIZE], i;
   int m = 100, a = 5, c = 13;
   x[0] = 11;
   for (i = 0; i < 100; i++)
   {
      x[i + 1] = (a * x[i] + c) % m;
   }
   printf("Generation of random numbers using linear congruential method:\n");
   for (i = 0; i < 100; i++)
   {
      printf("%d\t", x[i]);
   }
   return 0;
}
```

Output:

```
C:\csit\5_fifth sem\jonash sim

Generation of random numbers using linear congruential method:
11      68      53      78      3       28      53      78      3       28      53
78      3       28      53      78      3       28      53      78      3       28
53      78      3       28      53      78      3       28      53      78      3
28      53      78      3       28      53      78      3       28      53      78
3       28      53      78      3       28      53      78      3       28      53
78      3       28      53      78      328     53      78      3       28      53
78      3       28      53      78      3       28      53      78      328     53
78      3       28      53      78      3       28      53      78      3       28
53      78      328     53      78      3       28      53      78
--------------------------------
Process exited after 0.1167 seconds with return value 0
Press any key to continue . . .
```

Q.N. 2: WAP in C to generate 100 random numbers using Multiplicative Congruential Method where X_0=13, m =1000, a = 15 and c = 7.

Source Code:

```c
#include <stdio.h>
#define SIZE 100
int main()
{
   long x[SIZE];
   long m = 1000, a = 15, c = 7;
   int i;
   x[0] = 13;

   for (i = 0; i < SIZE - 1; i++)
   {
      x[i + 1] = (a * x[i] + c) % m;
   }
   printf("Generated 100 random numbers:\n\n");
   for (i = 0; i < SIZE; i++)
   {
      printf("%ld ", x[i]);
   }
   return 0;
}
```

Output:

```
C:\csit\5_fifth sem\jonash sim

Generated 100 random numbers:

13 202 37 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437
562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 56
2 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562
437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 43
7 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562
--------------------------------
Process exited after 0.09535 seconds with return value 0
Press any key to continue . . .
```

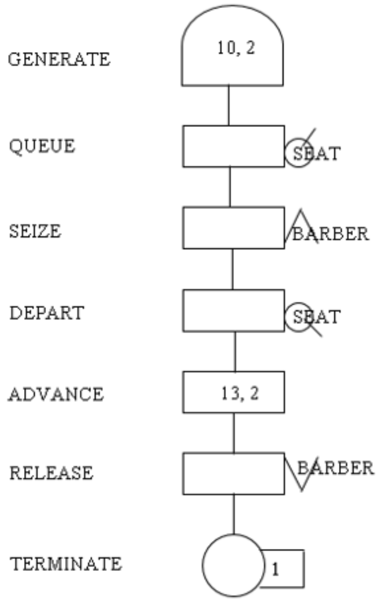
Madan Bhandari Memorial College

Lab No. 7

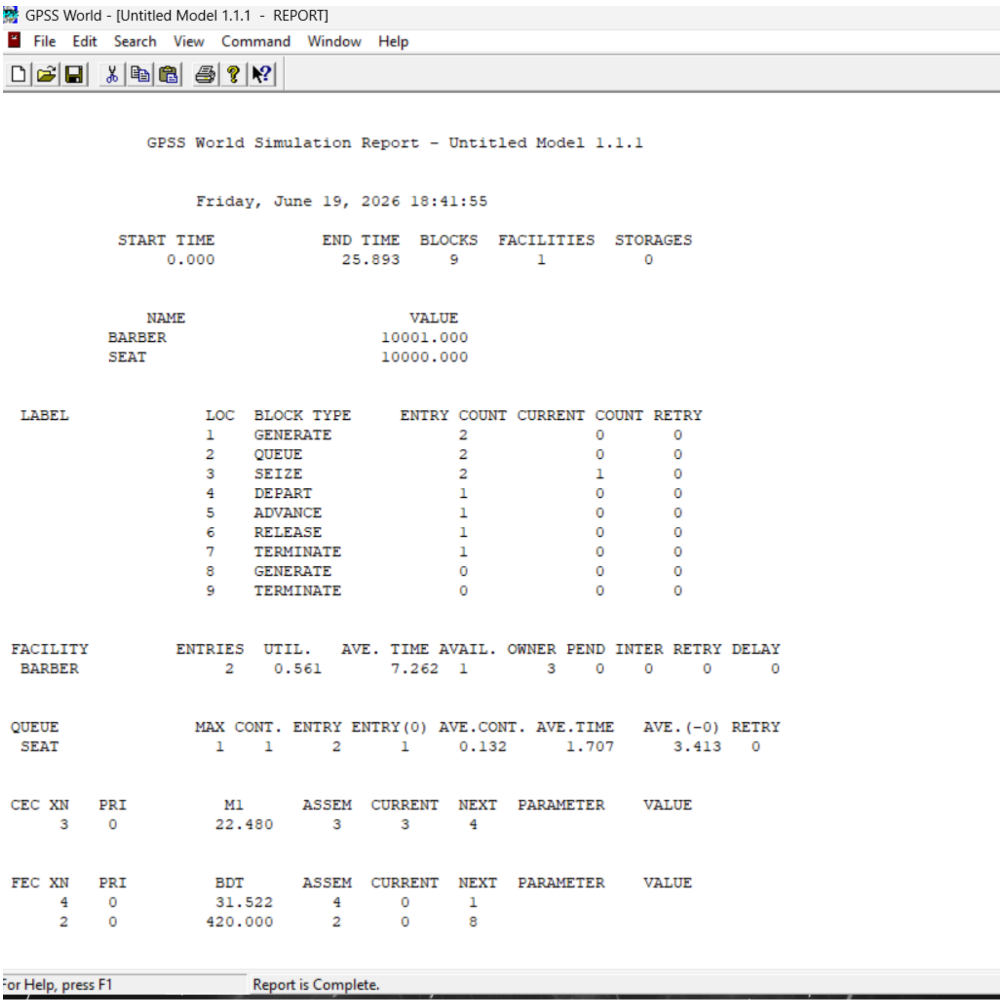
Create a GPSS model and program to simulate a barber shop for a day (9 a.m. to 4 p.m.), where a customer enters the shop every 10 ± 2 minutes and a barber takes 13 ± 2 minutes for a haircut.

Source Code:

```
GENERATE 10,2
QUEUE SEAT
SEIZE BARBER
DEPART SEAT
ADVANCE 13,2
RELEASE BARBER
TERMINATE 1
GENERATE 420
TERMINATE 1
START 1
```



Output:



GPSS World - [Untitled Model 1.1.1 - REPORT]

File Edit Search View Command Window Help

```
              GPSS World Simulation Report - Untitled Model 1.1.1


                   Friday, June 19, 2026 18:41:55

           START TIME           END TIME  BLOCKS  FACILITIES  STORAGES
                0.000             25.893     9         1          0


              NAME                       VALUE
          BARBER                       10001.000
          SEAT                         10000.000


 LABEL              LOC  BLOCK TYPE     ENTRY COUNT CURRENT COUNT RETRY
                    1    GENERATE             2             0       0
                    2    QUEUE                2             0       0
                    3    SEIZE                2             1       0
                    4    DEPART               1             0       0
                    5    ADVANCE              1             0       0
                    6    RELEASE              1             0       0
                    7    TERMINATE            1             0       0
                    8    GENERATE             0             0       0
                    9    TERMINATE            0             0       0


FACILITY         ENTRIES  UTIL.   AVE. TIME AVAIL. OWNER PEND INTER RETRY DELAY
 BARBER              2    0.561       7.262  1        3    0    0     0      0


QUEUE              MAX CONT. ENTRY ENTRY(0) AVE.CONT. AVE.TIME   AVE.(-0) RETRY
 SEAT                1    1     2      1     0.132     1.707      3.413   0


CEC XN   PRI          M1      ASSEM  CURRENT  NEXT  PARAMETER    VALUE
     3    0       22.480      3       3       4


FEC XN   PRI         BDT      ASSEM  CURRENT  NEXT  PARAMETER    VALUE
     4    0       31.522      4       0       1
     2    0      420.000      2       0       8
```

For Help, press F1 | Report is Complete.

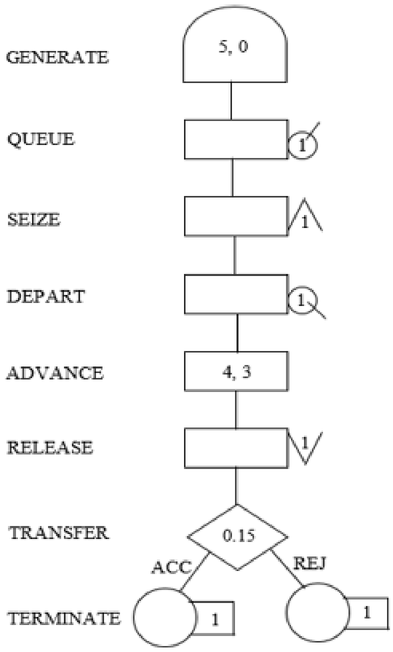
Madan Bhandari Memorial College

Lab No. 8

A machine tool in a manufacturing shop is turning out parts at the rate of every 5 minutes. When they are finished, the parts are sent to an inspector, who takes 4±3 minutes to examine each one and rejects 15% of the parts. Draw and explain a block diagram for it and write a GPSS program to simulate using the concept of FACILITY.

Source Code:

```
GENERATE 5,0
QUEUE INSQ
SEIZE INSPECTOR
DEPART INSQ
ADVANCE 4,3
RELEASE
INSPECTOR
TRANSFER
0.15,REJ,ACC
ACC TERMINATE 1
REJ TERMINATE 1
```



Output:

GPSS World - [Untitled Model 1.1.1 - REPORT]

File Edit Search View Command Window Help

```
              GPSS World Simulation Report - Untitled Model 1.1.1


                   Friday, June 19, 2026 18:47:20

           START TIME           END TIME  BLOCKS  FACILITIES  STORAGES
                0.000             11.286     9         1          0


              NAME                       VALUE
          ACC                              8.000
          INSPECTOR                    10001.000
          INSQ                         10000.000
          REJ                              9.000


 LABEL              LOC  BLOCK TYPE     ENTRY COUNT CURRENT COUNT RETRY
                    1    GENERATE             2             0       0
                    2    QUEUE                2             0       0
                    3    SEIZE                2             1       0
                    4    DEPART               1             0       0
                    5    ADVANCE              1             0       0
                    6    RELEASE              1             0       0
                    7    TRANSFER             1             0       0
ACC                 8    TERMINATE            0             0       0
REJ                 9    TERMINATE            1             0       0


FACILITY         ENTRIES  UTIL.   AVE. TIME AVAIL. OWNER PEND INTER RETRY DELAY
 INSPECTOR            2    0.557      3.143  1        2    0    0     0     0


QUEUE              MAX CONT. ENTRY ENTRY(0) AVE.CONT. AVE.TIME   AVE.(-0) RETRY
 INSQ                1    1      2      1     0.114      0.643      1.286   0


CEC XN   PRI          M1      ASSEM  CURRENT  NEXT  PARAMETER    VALUE
     2    0          10.000     2      3       4


FEC XN   PRI          BDT     ASSEM  CURRENT  NEXT  PARAMETER    VALUE
     3    0          15.000     3      0       1
```

Madan Bhandari Memorial College

Lab No. 9

A GENERATE block is used to represent the output of the machine by creating one transaction every five units of time. A QUEUE block places the transaction in the queue and SEIZE block allows a transaction to engage a facility if it is available. If more than one inspector is available, the transaction leaves the queue which is denoted by DEPART block and enters into ADVANCE block. An ADVANCE block with a mean of 4 and modifier of 3 is used to represent inspection. The time spent on inspection will therefore be any one of the values 1, 2, 3, 4, 5, 6 or 7, with equal probability given to each value. Upon completion of the inspection, RELEASE block allows a transaction to disengage the facility and transaction go to a TRANSFER block with a selection factor of 0.15, so that 85% of the parts go to the next location called ACC, to represent accepted parts and 15% go to another location called REJ to represent rejects. Both locations reached from the TRANSFER block are TERMINATE blocks.

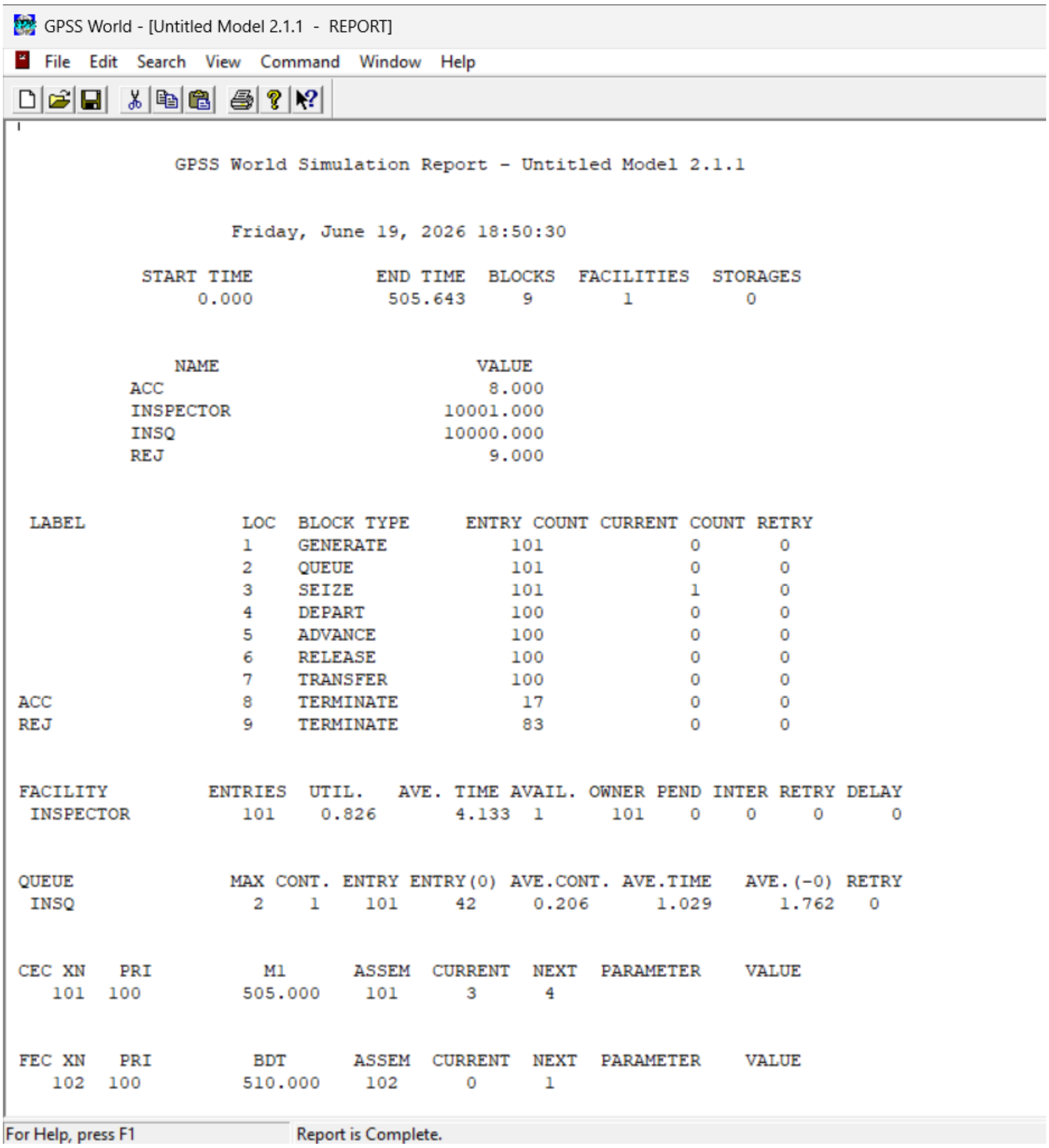
Source Code:

```
GENERATE 5,0,,,100
QUEUE INSQ
SEIZE INSPECTOR
DEPART INSQ
ADVANCE 4,3

RELEASE INSPECTOR
TRANSFER 0.15,REJ,ACC

ACC TERMINATE 1
REJ TERMINATE 1
START 100
```

Output:



```
GPSS World - [Untitled Model 2.1.1  -  REPORT]
File  Edit  Search  View  Command  Window  Help

              GPSS World Simulation Report - Untitled Model 2.1.1


                   Friday, June 19, 2026 18:50:30

           START TIME           END TIME  BLOCKS  FACILITIES  STORAGES
                0.000            505.643     9         1          0


              NAME                       VALUE
          ACC                              8.000
          INSPECTOR                    10001.000
          INSQ                         10000.000
          REJ                              9.000


 LABEL              LOC  BLOCK TYPE     ENTRY COUNT CURRENT COUNT RETRY
                    1    GENERATE           101             0       0
                    2    QUEUE              101             0       0
                    3    SEIZE              101             1       0
                    4    DEPART             100             0       0
                    5    ADVANCE            100             0       0
                    6    RELEASE            100             0       0
                    7    TRANSFER           100             0       0
ACC                 8    TERMINATE           17             0       0
REJ                 9    TERMINATE           83             0       0


FACILITY         ENTRIES  UTIL.   AVE. TIME AVAIL. OWNER PEND INTER RETRY DELAY
 INSPECTOR          101    0.826       4.133  1       101    0    0     0     0


QUEUE              MAX CONT. ENTRY ENTRY(0) AVE.CONT. AVE.TIME   AVE.(-0) RETRY
 INSQ                2    1    101     42      0.206      1.029      1.762   0


CEC XN   PRI          M1      ASSEM  CURRENT  NEXT  PARAMETER    VALUE
   101  100       505.000    101      3       4


FEC XN   PRI          BDT     ASSEM  CURRENT  NEXT  PARAMETER    VALUE
   102  100       510.000    102      0       1

For Help, press F1          Report is Complete.
```

Lab No. 10

Parts are being made at the rate of one every 10 minutes. They are of two types; A and B. Mixed randomly with 10% being type B. Separate inspectors inspect each type. Inspection time for A is 6±2 minutes and for B is 10±2 minutes. Both inspectors reject 10% of parts. Simulate for 100 parts.

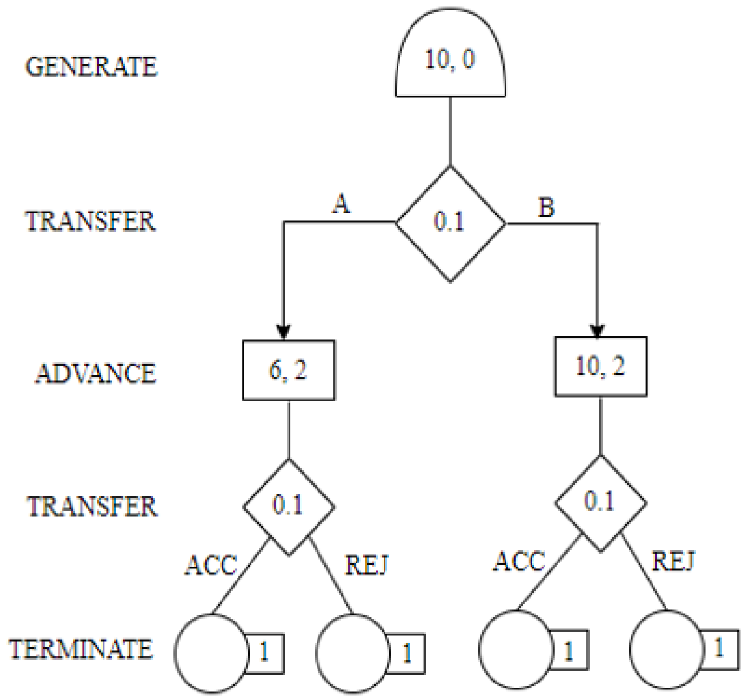
Source Code:

```
GENERATE 10,,,100
TRANSFER .10,TYPEB,TYPEA
TYPEA QUEUE AQUEUE
    SEIZE INSPA
    DEPART AQUEUE
    ADVANCE 6,2
    RELEASE INSPA
    TRANSFER .10,REJA,ACCA

ACCA  TERMINATE 1
REJA  TERMINATE 1

TYPEB QUEUE BQUEUE
    SEIZE INSPB
    DEPART BQUEUE
    ADVANCE 10,2
    RELEASE INSPB
    TRANSFER .10,REJB,ACCB

ACCB  TERMINATE 1
REJB  TERMINATE 1
START 100
```



Output:

GPSS World - [Untitled Model 3.1.1 - REPORT]

File Edit Search View Command Window Help

```
              GPSS World Simulation Report - Untitled Model 3.1.1


                   Friday, June 19, 2026 18:54:20

           START TIME           END TIME  BLOCKS  FACILITIES  STORAGES
                0.000           1008.573    18        2          0


              NAME                       VALUE
          ACCA                             9.000
          ACCB                            17.000
          AQUEUE                       10002.000
          BQUEUE                       10000.000
          INSPA                        10003.000
          INSPB                        10001.000
          REJA                            10.000
          REJB                            18.000
          TYPEA                            3.000
          TYPEB                           11.000


 LABEL              LOC  BLOCK TYPE     ENTRY COUNT CURRENT COUNT RETRY
                      1  GENERATE           100             0       0
                      2  TRANSFER           100             0       0
TYPEA                 3  QUEUE               11             0       0
                      4  SEIZE               11             0       0
                      5  DEPART              11             0       0
                      6  ADVANCE             11             0       0
                      7  RELEASE             11             0       0
                      8  TRANSFER            11             0       0
ACCA                  9  TERMINATE            2             0       0
REJA                 10  TERMINATE            9             0       0
TYPEB                11  QUEUE               89             0       0
                     12  SEIZE               89             0       0
                     13  DEPART              89             0       0
                     14  ADVANCE             89             0       0
                     15  RELEASE             89             0       0
                     16  TRANSFER            89             0       0
ACCB                 17  TERMINATE           11             0       0
REJB                 18  TERMINATE           78             0       0


FACILITY         ENTRIES  UTIL.   AVE. TIME AVAIL. OWNER PEND INTER RETRY DELAY
 INSPB              89    0.871       9.871  1        0    0    0     0     0
 INSPA              11    0.075       6.849  1        0    0    0     0     0


QUEUE              MAX CONT. ENTRY ENTRY(0) AVE.CONT. AVE.TIME   AVE.(-0) RETRY
 BQUEUE              1    0     89     37     0.115     1.300      2.225   0
 AQUEUE              1    0     11     11     0.000     0.000      0.000   0
```

For Help, press F1 Report is Complete.